# Mappings and Elementary Functions

Mathematics for Engineers I · Module 62765 · Chapter 3

From Fundamentals to AI Apocalypse · personal study guide

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Made by Thanate Amornjiraporn (Focus)

This chapter builds the language of functions: how to describe mappings, test whether they have inverses, and combine them. It then applies that language to polynomials, trigonometric and exponential functions, and polar coordinates.

## Start from memory

Try these before reading. Write an answer even if you are uncertain; the gap tells you what to look for.

1. What must be true for a rule from X to Y to be a mapping?
2. How do injective and surjective describe a mapping?
3. When does an inverse function exist?
4. In what order do you apply the functions in $g \circ f$?
5. What does the discriminant tell you about a quadratic's real roots?
6. How do polar coordinates convert to Cartesian coordinates?

*Answers: see "Start from memory" under Answers at the end.*

## How to use this guide

Attempt before reading each worked solution. Then hide the guide and retrieve the method on a later day.

1. **Day 0:** Do the cold start. Read §§3.1–3.2 and draw a mapping diagram that distinguishes domain, codomain and image.
2. **Day 1:** Read §§3.3–3.4. Re-solve the inverse and composition examples from a blank page, then try mixed practice 1–8.
3. **Day 3:** Study the polynomial and trigonometric parts of §3.5. Rebuild the quadratic formula derivation, then solve practice 9–16 without notes.
4. **About day 7:** Study exponentials, logarithms and §3.6. Shuffle practice 17–24 and explain aloud which domain or quadrant condition matters.
5. **About day 14:** Rebuild all listed derivations and retry every marked question. Complete the self-check only with evidence from unaided work.
6. **Before the exam:** Do both exam problems under time limits, then review only the skills your evidence shows are weak.

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## How the chapter fits together

Mappings give a precise language for inputs and outputs. Injectivity supports inverses, compatible domains support composition, and elementary functions provide the examples used in coordinate changes.

### Measured · examples and coordinates

- **Domain, codomain and image** §3.1
  $f:X\to Y$; $\mathrm{Im}(f)\subseteq Y$
  Rests on: assigning exactly one output to each input.
- **Elementary functions** §3.5
  $P(x)$, $\sin x$, $e^x$, $\ln x$
  Rests on: domain, rule and range being stated together.

### Defined · structure and operations

- **Injective, surjective, bijective** §3.2
  one-to-one; onto; both
  Rests on: comparing the image with the codomain.
- **Composition** §3.4
  $(g\circ f)(x)=g(f(x))$
  Rests on: $\mathrm{Im}(f)$ fitting inside $D(g)$.
- **Polar coordinates** §3.6
  $x=r\cos\varphi$, $y=r\sin\varphi$
  Rests on: sine and cosine on the unit circle.

### Derived · reversible and reusable results

- **Inverse function** §3.3
  $f^{-1}\circ f=\mathrm{id}_X$
  Rests on: injectivity, so each reached output has one input.
- **Quadratic roots** §3.5
  $\Delta=b^2-4ac$
  Rests on: completing the square and checking $\Delta$'s sign.
- **Cartesian conversion** §3.6
  $r=\sqrt{(x^2+y^2)}$
  Rests on: right-triangle length and a quadrant-aware angle.

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### Choose a tool

| If the task asks... | Choose this tool |
|---|---|
| Does a rule define a function? | Check that each domain input has exactly one output. |
| Can you reverse the mapping? | Test injectivity, then solve y=f(x) for x. |
| What is f after g? | Write the composition order and check the intermediate domain. |
| How many real roots can a polynomial have? | Use degree for a bound; use $\Delta$ for a quadratic. |
| Where is an angle or point? | Use the unit circle or quadrant-aware polar conversion. |

## 3.1 The Graph of a Mapping

A function is more than a formula: its domain and codomain are part of its definition. This lets you ask exactly which outputs occur and which inputs lead to a chosen set of outputs.

**Mapping and image**

$$f : X \to Y; \quad \mathrm{Im}(f) = \{f(x) : x \in X\} \subseteq Y$$

X is the domain; Y is the codomain; Im(f) is the image or range. Every $x \in X$ must be assigned exactly one value $f(x) \in Y$.

**Image, preimage and graph**

$$f(A) = \{f(x) : x \in A\}; \quad f^{-1}(B) = \{x \in X : f(x) \in B\}; \quad \mathrm{Gr}(f) = \{(x, y) \in X \times Y : y = f(x)\}$$

$A \subseteq X$ and $B \subseteq Y$. Preimage notation $f^{-1}(B)$ is defined even when f has no inverse function.

A real-valued function has a graph in the coordinate plane. Each vertical line must meet the graph at most once: two y-values for one x would violate the mapping rule. A horizontal line may meet it more than once; that tells you the function is not injective, not that it fails to be a function.

**Worked example · image and preimage**

Let $f:\mathbb{R}\to\mathbb{R}$ be $f(x)=x^2$, and let $A=\{-2,0,3\}$, $B=\{4\}$.

1. Apply the rule to each member of A: $f(A)=\{4,0,9\}$.
2. To find $f^{-1}(B)$, solve $x^2=4$ over the stated domain $\mathbb{R}$. Both $x=-2$ and $x=2$ work, so $f^{-1}(\{4\})=\{-2,2\}$.
3. The image is $[0,\infty)$, which is smaller than the codomain $\mathbb{R}$. Check: a square cannot be negative.

**Trap:** $f^{-1}(B)$ means "inputs that land in B"; it does not mean "apply the inverse function". The inverse function exists only if f is injective.

**Recall check · answer before opening**

1. What three sets are involved in $f:X\to Y$?
2. Can a preimage be defined when f is not injective?

*Answers: see "3.1 The Graph of a Mapping" under Answers at the end.*

## 3.2 Types of Mappings

Injectivity and surjectivity answer different questions. Together they determine whether every output has exactly one input.

**Injective**

$$x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$$

Distinct inputs have distinct outputs; equivalently, $f(x_1)=f(x_2)$ implies $x_1=x_2$.

**Surjective and bijective**

$$f \text{ is onto } Y \text{ iff } \mathrm{Im}(f) = Y; \quad \text{bijective iff injective and surjective}$$

Surjectivity depends on the chosen codomain. A function is always surjective when its codomain is restricted to its image.

For $f:\mathbb{R}\to\mathbb{R}$, $f(x)=x^3$ is strictly increasing and reaches every real value, so it is bijective. In contrast, $x^2$ maps 1 and $-1$ to the same output and never reaches negative values: it is neither injective nor surjective onto $\mathbb{R}$. Restricting it to $f:[0,\infty)\to[0,\infty)$ makes it bijective.

**Worked example · the codomain changes surjectivity**

Consider $f(x)=x^2$ with domain $[0,\infty)$.

1. If the codomain is $\mathbb{R}$, the mapping is injective but not surjective, because no negative real is a square.
2. If the codomain is $[0,\infty)$, every codomain value $y$ has the preimage $\sqrt{y}$. Thus it is surjective too.
3. So $f:[0,\infty)\to[0,\infty)$ is bijective. Check: each $y\geq 0$ has exactly one non-negative square root.

**Memory hook:** injective asks "does any output have two inputs?"; surjective asks "does any codomain value have no input?". State the codomain before answering.

**Recall check · answer before opening**

1. What is the difference between injective and surjective?
2. Why is $x\mapsto x^2$ not onto $\mathbb{R}$?

*Answers: see "3.2 Types of Mappings" under Answers at the end.*

## 3.3 Inverse Function

An inverse undoes a mapping. It can do that consistently only when an output identifies one input.

**Inverse function**

$$f : X \to Y \text{ injective } \Rightarrow f^{-1} : \mathrm{Im}(f) \to X$$

For $y \in \mathrm{Im}(f)$, $f^{-1}(y)$ is the unique $x \in X$ satisfying $f(x)=y$. The inverse domain is $\mathrm{Im}(f)$, not necessarily all of Y.

**Inverse identities**

$$f^{-1} \circ f = \mathrm{id}_X; \quad f \circ f^{-1} = \mathrm{id}_{\mathrm{Im}(f)}$$

The domain and image switch. For real-valued functions, the two graphs reflect across $y=x$.

To find an inverse, write $y=f(x)$, solve for x, then rename the output variable. Keep the domain restrictions: the inverse is defined only for values in the original image. Check both compositions when the domains allow it.

**Worked example · solve for the inverse**

Let $f:\mathbb{R}\to\mathbb{R}$, $f(x)=2x+3$.

1. Set $y=2x+3$ and solve: $x=(y-3)/2$.
2. Therefore $f^{-1}(y)=(y-3)/2$, with domain $\mathbb{R}$.
3. Check: $f^{-1}(f(x))=((2x+3)-3)/2=x$. Also $f(f^{-1}(y))=2((y-3)/2)+3=y$.

**Trap:** $f^{-1}(x)$ is not $1/f(x)$. It means the inverse mapping. If f is not injective, solving $y=f(x)$ may give multiple x-values and no inverse function.

**Recall check · answer before opening**

1. What is the domain of $f^{-1}$?
2. How can you check a proposed inverse?

*Answers: see "3.3 Inverse Function" under Answers at the end.*

## 3.4 Composition

Composition chains mappings. The output of the first mapping must be a valid input to the next.

**Composition**

$$\text{If } f : X \to Y,\ g : D(g) \subseteq Y \to Z, \text{ and } \mathrm{Im}(f) \subseteq D(g), \text{ then } (g \circ f)(x) = g(f(x))$$

Read right to left: apply f first, then g. The containment condition guarantees the second step is defined for every first-step output.

**Composition laws**

$$h \circ (g \circ f) = (h \circ g) \circ f; \quad \text{in general } g \circ f \neq f \circ g$$

Composition is associative when the domains fit, but order usually matters. For a bijection, $f^{-1} \circ f = \mathrm{id}_X$.

**Worked example · order and domain**

Let $f:\mathbb{R}\to\mathbb{R}$ be $f(x)=2x+1$ and $g:\mathbb{R}\to\mathbb{R}$ be $g(x)=x^2$.

1. For $g\circ f$, apply f then square: $(g\circ f)(x)=(2x+1)^2$.
2. For $f\circ g$, square then apply f: $(f\circ g)(x)=2x^2+1$.
3. At $x=1$ these give 9 and 3, so the two compositions differ. This numerical check catches a reversed order.

**Trap:** do not apply the left-hand function first. Write the nested expression $g(f(x))$ before simplifying, and check that f's image lies in g's domain.

**Recall check · answer before opening**

1. In $g\circ f$, which mapping acts first?
2. What must be true for $g\circ f$ to be defined on all of X?

*Answers: see "3.4 Composition" under Answers at the end.*

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## 3.5 Elementary Functions

This section collects important real-valued mappings. Their domain and range determine where inverse functions and formulas make sense.

### Real polynomials

**Polynomial and roots**

$$P(x) = a_n x^n + \ldots + a_1 x + a_0, \quad a_n \neq 0; \quad P(x_0) = 0 \text{ means } x_0 \text{ is a root}$$

The degree is n and the domain is $\mathbb{R}$. A degree-n polynomial has at most n real roots.

**Quadratic formula and discriminant**

$$\Delta = b^2 - 4ac; \quad x = (-b \pm \sqrt{\Delta})/(2a), \quad ax^2 + bx + c = 0,\ a \neq 0$$

$\Delta>0$ gives two distinct real roots; $\Delta=0$ gives one repeated root; $\Delta<0$ gives no real roots. When $\Delta\geq 0$, the roots factor the polynomial as $a(x-x_1)(x-x_2)$.

Completing the square reveals why $\Delta$ controls the roots. Multiply the equation by 4a, add $b^2$, and rewrite the left side as $(2ax+b)^2$. Taking square roots creates a real solution only if $\Delta\geq 0$.

**Worked example · solve a quadratic**

Solve $x^2-5x+6=0$.

1. Here $a=1$, $b=-5$, $c=6$, so $\Delta=(-5)^2-4\cdot 1\cdot 6=1$.
2. The roots are $(5\pm 1)/2$, hence $x=2$ or $x=3$.
3. Check by factoring: $x^2-5x+6=(x-2)(x-3)$. Substitution gives zero at both roots.

The notes mention general root formulas for degrees 3 and 4, but no general formula for every polynomial of degree at least 5. This does not prevent a particular higher-degree polynomial from factoring or having roots found by another method.

**Sign check:** for $\Delta<0$ there are no real roots, but the polynomial still has a real graph. "No real roots" does not mean "no roots in every number system."

## Trigonometric functions

### Unit circle and radians

$$P = (\cos x, \sin x); \quad \tan x = \sin x / \cos x$$

Angles are in radians. One full turn is $2\pi$ radians $= 360°$. On a right triangle, sin x is opposite/hypotenuse and cos x is adjacent/hypotenuse for acute x.

| Degrees | 0° | 30° | 45° | 60° | 90° | 180° | 360° |
|---|---|---|---|---|---|---|---|
| Radians | 0 | $\pi/6$ | $\pi/4$ | $\pi/3$ | $\pi/2$ | $\pi$ | $2\pi$ |

### Core trigonometric identities

$$\sin(-x) = -\sin x; \quad \cos(-x) = \cos x; \quad \sin^2 x + \cos^2 x = 1$$
$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$
$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$

sin and cos have period $2\pi$. Their zeros are sin x=0 at $x=k\pi$ and cos x=0 at $x=\pi/2+k\pi$, $k\in\mathbb{Z}$.

### Double-angle and square identities

$$\sin(2x) = 2\sin x \cos x; \quad \cos(2x) = 1 - 2\sin^2 x$$
$$\sin^2 x = (1 - \cos 2x)/2; \quad \cos^2 x = (1 + \cos 2x)/2$$

These follow from the addition formulas and $\sin^2 x+\cos^2 x=1$.

### Difference-to-product identities

$$\sin x - \sin y = 2\cos((x+y)/2)\sin((x-y)/2)$$
$$\cos x - \cos y = 2\sin((x+y)/2)\sin((y-x)/2)$$

The second factor uses $(y-x)/2$; reversing its sign would reverse the whole result.

**Worked example · use an addition formula**

Find $\sin(\pi/3+\pi/6)$.

1. Use $\sin(x+y)=\sin x \cos y+\cos x \sin y$.
2. Substitute the special-angle values: $(\sqrt{3}/2)(\sqrt{3}/2)+(1/2)(1/2)=3/4+1/4=1$.
3. The result matches $\sin(\pi/2)=1$, a useful unit-circle check.

For inverse functions, restrict sine to $[-\pi/2,\pi/2]$, cosine to $[0,\pi]$, and tangent to $(-\pi/2,\pi/2)$. Then arcsin and arccos have domain $[-1,1]$, while arctan has domain $\mathbb{R}$. $\tan x$ is undefined when $\cos x=0$, that is $x=\pi/2+k\pi$.

**Tangent identities**

$$\tan(-x) = -\tan x; \quad \tan(x \pm y) = (\tan x \pm \tan y)/(1 \mp \tan x \tan y)$$
$$\tan(2x) = 2\tan x/(1 - \tan^2 x)$$

Use only where each tangent and denominator is defined. These follow by dividing the sine and cosine addition formulas.

**Trap:** inverse trigonometric functions return principal angles, not every angle with that sine or cosine. Check the interval and quadrant before giving a general solution.

## Exponential and logarithmic functions

**Exponential and natural logarithm**

$$e^{x+y} = e^x e^y; \quad \ln(e^x) = x; \quad e^{\ln y} = y \text{ for } y > 0$$
$$\ln(xy) = \ln x + \ln y \text{ for } x, y > 0; \quad \ln 1 = 0$$

$e^x$ is strictly increasing, positive, and has range $(0,\infty)$. ln is its inverse, with domain $(0,\infty)$.

**Base a and real powers**

$$a^x = e^{x \ln a} \ (a > 0); \quad \log_a x = \ln x/\ln a \ (a > 0,\ a \neq 1,\ x > 0)$$
$$x^\alpha = e^{\alpha \ln x} \text{ for } x > 0,\ \alpha \in \mathbb{R}$$

For $0<a<1$, $a^x$ decreases; for $a>1$ it increases. The notes define arbitrary real powers $x^\alpha$ only for positive x.

**Worked example · solve an exponential equation**

Solve $5^x=125$.

1. Write $125=5^3$, so $5^x=5^3$.
2. Because the base 5 is positive and not 1, the exponential is injective. Therefore $x=3$.
3. Check: $5^3=125$ and x is allowed in the domain $\mathbb{R}$.

**Trap:** ln x is defined for $x>0$ only. Also, $\log_a$ requires $a>0$ and $a\neq1$; $a=1$ would give a constant function with no inverse.

**Recall check · answer before opening**

1. What does the sign of a quadratic's discriminant tell you?
2. On which interval is sin restricted to define arcsin?
3. What are the domain and range of ln?

*Answers: see "3.5 Elementary Functions" under Answers at the end.*

## 3.6 Polar Coordinates

Cartesian coordinates locate a point by horizontal and vertical displacement. Polar coordinates use distance from the origin and angle from the positive x-axis.

**Polar to Cartesian**

$$x = r\cos\varphi; \quad y = r\sin\varphi$$

Here $r \geq 0$ and $\varphi$ is measured in radians from the positive x-axis. The point is $(x,y)=(r\cos\varphi, r\sin\varphi)$.

**Cartesian to polar**

$$r = \sqrt{(x^2 + y^2)}; \quad \tan\varphi = y/x \text{ when } x \neq 0$$

Use the signs of x and y to select the quadrant. For $0 \leq \varphi < 2\pi$: use $\arctan(y/x)$ when $x>0, y\geq 0$; add $2\pi$ when $x>0, y<0$; add $\pi$ when $x<0$; take $\pi/2$ when $x=0, y>0$ and $3\pi/2$ when $x=0, y<0$. The origin has $r=0$ and no unique angle.

With $r \geq 0$ and unrestricted angle, polar coordinates are not unique: adding $2\pi$ does not move the point, and the origin has every angle. Restricting to $r>0$ and $0 \leq \varphi < 2\pi$ gives one polar pair for every non-origin point. The origin must be handled separately if it is included.

**Worked example · convert a point and a region**

1. Convert $(x,y)=(-1,1)$. The radius is $r=\sqrt{(1+1)}=\sqrt{2}$. The point is in quadrant II, so $\varphi=3\pi/4$. Thus $(-1,1)=(\sqrt{2}, 3\pi/4)$.
2. Convert the region $0 \leq r \leq 2$, $0 \leq \varphi \leq \pi/4$. Since $r^2=x^2+y^2$, it lies inside the circle $x^2+y^2 \leq 4$. The angle range is the first-quadrant wedge between $y=0$ and $y=x$, so $0 \leq y \leq x$.
3. Check a boundary point: $r=2$, $\varphi=\pi/4$ gives $x=y=\sqrt{2}$ and $x^2+y^2=4$.

**Sign check:** $\arctan(y/x)$ alone returns an angle only in its principal range. Use the signs of both coordinates; when $x=0$, use the vertical-axis cases, and at the origin the angle is undefined.

**Recall check · answer before opening**

1. What are x and y in terms of r and $\varphi$?
2. Why must you check the quadrant when finding $\varphi$?

*Answers: see "3.6 Polar Coordinates" under Answers at the end.*

## Formula map and checks

This chapter has no SI-valued quantities; domains, ranges and angle units (radians) take the place of physical units.

| Topic | Relation | Check |
|---|---|---|
| Image and preimage | $\mathrm{Im}(f)=\{f(x):x\in X\}$; $f^{-1}(B)=\{x:f(x)\in B\}$ | Keep image values in the codomain; preimage is not an inverse function. |
| Types | Injective: equal outputs imply equal inputs; onto: $\mathrm{Im}(f)=Y$ | State the codomain before deciding "onto". |
| Inverse | $f^{-1}:\mathrm{Im}(f)\to X$; $f^{-1}\circ f=\mathrm{id}_X$ | Only for an injective mapping. |
| Composition | $(g\circ f)(x)=g(f(x))$ | f first; verify $\mathrm{Im}(f)\subseteq D(g)$. |
| Quadratic | $\Delta=b^2-4ac$; $x=(-b\pm\sqrt{\Delta})/(2a)$ | Use real-root cases by the sign of $\Delta$; require $a\neq 0$. |
| Trigonometry | $\sin^2x+\cos^2x=1$; $\sin(x\pm y)=\sin x\cos y\pm\cos x\sin y$; $\cos(x\pm y)=\cos x\cos y\mp\sin x\sin y$; $\sin(2x)=2\sin x\cos x$; $\sin x-\sin y=2\cos((x+y)/2)\sin((x-y)/2)$; $\tan(x\pm y)=(\tan x\pm\tan y)/(1\mp\tan x\tan y)$ | Angles are radians; exclude cosine or denominators equal to zero. |
| Exponential/logarithm | $e^{x+y}=e^xe^y$; $\ln(xy)=\ln x+\ln y$; $\log_a x=\ln x/\ln a$ | $x,y>0$ and $a>0$, $a\neq 1$ for $\log_a$. |
| Polar conversion | $x=r\cos\varphi$; $y=r\sin\varphi$; $r=\sqrt{(x^2+y^2)}$ | Angles are radians; check quadrant; origin has no unique angle. |

## Derive it cold

Rebuild these on blank paper before opening the worked examples in the relevant sections.

1. Show that the graph of a real-valued function meets each vertical line at most once. §3.1
2. Derive the range of $x\mapsto x^2$ on $\mathbb{R}$ and show why it is not injective there. §3.2
3. Starting from $y=f(x)$, solve for $x$ to obtain the inverse of a one-to-one affine function. §3.3
4. Compute both $g\circ f$ and $f\circ g$ for $f(x)=2x+1$, $g(x)=x^2$, and compare. §3.4
5. Complete the square in $ax^2+bx+c=0$ and derive the quadratic formula. §3.5
6. Derive the double-angle formulas from the sine and cosine addition formulas. §3.5
7. Derive $x=r\cos\varphi$ and $y=r\sin\varphi$ from a right triangle, then recover $r$ by Pythagoras. §3.6

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## Explain it

Answer aloud in complete sentences. Then compare with the model answer. (Retrieval and explanatory questioning [MIS, AMN, NEW].)

1. **Why** is the codomain part of a function's definition?
2. **When** can a mapping be surjective after changing only its codomain?
3. **Why** does an inverse function require injectivity?
4. **When** is $g \circ f$ defined for every x in X?
5. **Why** does the discriminant determine the number of real quadratic roots?
6. **Why** must inverse trigonometric functions use restricted intervals?
7. **When** is $\log_a x$ defined, and why can a not equal 1?
8. **Why** do polar coordinates need a chosen angle interval?

*Answers: see "Explain it" under Answers at the end.*

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## Mixed practice

Try all 24 problems before opening the answer key. The topics are interleaved, so identify the method before calculating. (Desirable difficulty and varied retrieval [MIS, AMN].)

1. Let $f:\mathbb{Z}\to\mathbb{Z}$ be $f(x)=|x|$. Find $\mathrm{Im}(f)$ and $f^{-1}(\{3\})$.
2. Find the roots of $x^2+2x-8=0$.
3. Is $f:\mathbb{R}\to\mathbb{R}$, $f(x)=x^3$, injective, surjective, both or neither?
4. Convert $(x,y)=(-1,-\sqrt{3})$ to polar coordinates with $0\le\varphi<2\pi$.
5. Find the inverse of $f(x)=4x-7$ on $\mathbb{R}$.
6. For $f(x)=3x-2$ and $g(x)=x+4$, find $g\circ f$ and $f\circ g$.
7. For $f:\mathbb{N}\to\mathbb{N}$, $f(n)=n^2$, find $f(\{1,3,5\})$ and state its codomain image relationship.
8. Convert $150^\circ$ to radians and give $\cos(150^\circ)$.
9. Make $x\mapsto x^2$ bijective by choosing a domain and codomain restriction.
10. Solve $2^x=16$ and state which property makes your step valid.
11. Find the inverse of $f(x)=5x+10$ and verify $f^{-1}(f(2))$.
12. Let $f:\mathbb{R}\to[0,\infty)$, $f(x)=x^2$ and $g:[0,\infty)\to\mathbb{R}$, $g(t)=\sqrt{t}$. Find $g\circ f$.
13. Does the relation $y^2=x$ define $y$ as a function of $x$ on $\mathbb{R}$? Explain using one input.
14. Use $\sin(2x)=2\sin x\cos x$ to find $\sin(\pi/3)$.
15. How many real roots does $3x^2+2x+5=0$ have? Compute $\Delta$.
16. Convert $(r,\varphi)=(3,\pi/6)$ to Cartesian coordinates.
17. For $f(x)=x^2$ on $\mathbb{R}$, find $f^{-1}([1,4])$.
18. Find the degree of $P(x)=-2x^5+x^2-7$ and the greatest possible number of distinct real roots.
19. On which interval is cosine one-to-one in the notes, and what is the range on that interval?
20. Let $f(x)=x+1$, $g(x)=2x$ and $h(x)=x^2$. Compute $h\circ(g\circ f)(x)$.
21. For which real $x$ is $\tan x$ undefined?
22. Simplify $\ln(e^3)+\ln(e^2)$.
23. Convert $(x,y)=(0,-5)$ to polar coordinates with $0\le\varphi<2\pi$.
24. For $f:X\to Y$, if $\mathrm{Im}(f)=Y$ and $f$ is injective, what type of mapping is $f$?

*Answers: see "Mixed practice" under Answers at the end.*

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## Exam-style problems

Allow about 20 minutes for each. Work every part before checking the answer. Afterward, note where you lost a step and revisit the cited sections.

### 1 · Inverses, image and composition

Let $f:[0,\infty)\to[0,\infty)$, $f(x)=x^2$, and $g:[0,\infty)\to\mathbb{R}$, $g(t)=t-1$.

1. State whether f is injective and surjective, with reasons.
2. Find $f^{-1}$ and verify one inverse composition.
3. Find $g\circ f$, its image, and its domain.
4. Is $f\circ g$ defined on all of g's domain? Explain.

*Answers: see "Exam-style problems" under Answers at the end.*

### 2 · Quadratic, trigonometric and polar tools

Consider $P(x)=x^2-2x-3$ and the point with polar coordinates $(r,\varphi)=(4,\pi/3)$.

1. Use the discriminant to find and classify the roots of P.
2. Write P in factored form and verify one root.
3. Convert $(r,\varphi)$ to Cartesian coordinates.
4. Find $\sin(\pi/3+\pi/6)$ using an addition theorem, and explain why the result is consistent with the unit circle.

*Answers: see "Exam-style problems" under Answers at the end.*

## Slips in the notes

This is a source-checked issue. Printed page numbers refer to the lecture notes.

1. **Printed page 27, composition label:** the sentence says the composition "$f\circ g$" is defined, but the condition and displayed formula define "$g\circ f$", with $(g\circ f)(x)=g(f(x))$. The latter is the consistent statement: if $f:X\to Y$, $g:D(g)\subseteq Y\to Z$ and $\mathrm{Im}(f)\subseteq D(g)$, then $g\circ f:X\to Z$ is defined.

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## Self-check

Tick a skill only in a later session, at least one day after learning it, when you can show it without notes. Use the problem numbers as evidence.

- ☐ Decide whether a rule defines a mapping and identify its domain and codomain. (13, 24)
- ☐ Find images and preimages without confusing preimage with inverse. (1, 7, 17)
- ☐ Test injectivity, surjectivity and bijectivity for a stated codomain. (3, 9, 24)
- ☐ Find an inverse and verify it by composition. (5, 11, 19)
- ☐ Compose mappings in the right order and check domains. (6, 12, 20)
- ☐ Use polynomial degree to bound the number of real roots. (18)
- ☐ Use the discriminant and quadratic formula correctly. (2, 15)
- ☐ Convert common degree measures to radians. (8)
- ☐ Apply sine, cosine and tangent identities with domain restrictions. (14, 21)
- ☐ State the principal domains and ranges of inverse trigonometric functions. (19)
- ☐ Solve exponential equations using injectivity. (10)
- ☐ Use logarithms only on their valid domains. (22)
- ☐ Convert polar coordinates to Cartesian coordinates. (16)
- ☐ Convert Cartesian coordinates with the correct quadrant and origin case. (4, 23)
- ☐ Rebuild the listed derivations without notes.
- ☐ Solve both multi-part problems and justify every step.

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# Answers

## Start from memory

1. Every x in the domain must be assigned exactly one value in the codomain. §3.1
2. Injective means different inputs have different outputs; surjective means every codomain value is reached. §3.2
3. When the mapping is injective; then it is bijective from its domain to its image. §3.3
4. Apply f first, then g: $(g\circ f)(x)=g(f(x))$. §3.4
5. It distinguishes two, one repeated, or no real roots according as $\Delta$ is positive, zero, or negative. §3.5
6. $x=r\cos\varphi$ and $y=r\sin\varphi$. §3.6

## 3.1 The Graph of a Mapping

1. The domain X, codomain Y, and image $\mathrm{Im}(f)\subseteq Y$. §3.1
2. Yes. $f^{-1}(B)$ is the set of domain inputs whose outputs lie in B, whether or not f has an inverse function. §3.1

## 3.2 Types of Mappings

1. Injective prevents two distinct inputs sharing an output; surjective requires every codomain value to occur. §3.2
2. Its outputs are non-negative, so negative real numbers in the codomain have no preimage. §3.2

## 3.3 Inverse Function

1. The image Im(f) of the original mapping. §3.3
2. Compose in both directions on their stated domains and check that each composition is the identity. §3.3

## 3.4 Composition

1. f acts first, then g. §3.4
2. Every output of f must lie in the domain of g: $\mathrm{Im}(f)\subseteq D(g)$. §3.4

## 3.5 Elementary Functions

1. Positive: two distinct real roots; zero: one repeated real root; negative: no real roots. §3.5

$-\pi/2,\pi/2$ , where sine is one-to-one and has range $[-1,1]$. §3.5

2. Its domain is $(0,\infty)$ and its range is $\mathbb{R}$. §3.5

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## 3.6 Polar Coordinates

1. $x=r\cos\varphi$ and $y=r\sin\varphi$. §3.6
2. Because $\arctan(y/x)$ alone cannot distinguish opposite quadrants and division by x fails when $x=0$. §3.6

## Explain it

1. The codomain specifies which outputs count as possible and therefore determines whether the mapping is onto. The same rule can be surjective for one codomain and not another.
2. When the new codomain is the image of the mapping. Every image value is reached by definition.
3. If two inputs share one output, reversing that output would require choosing between them. Injectivity makes the input unique.
4. When every output of f lies in the domain of g, so $\mathrm{Im}(f)\subseteq D(g)$.
5. Completing the square yields $(2ax+b)^2=\Delta$. A real square root exists only for $\Delta\geq 0$; positive $\Delta$ gives two signs, zero gives one repeated value.
6. Sine, cosine and tangent repeat values, so they are not one-to-one on their full domains. Restricting to a principal interval makes each inverse return one defined angle.
7. x must be positive, and a must be positive but not 1. Otherwise ln x is not real, or $a^t$ is not invertible because it is constant at 1.
8. Without a restriction, angles differing by $2\pi$ describe the same point. The origin has no unique angle even after an interval is selected.

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## Mixed practice

1. $\mathrm{Im}(f)=\mathbb{N}_0$; the inputs with $|x|=3$ are $\{-3,3\}$. §3.1
2. $\Delta=4+32=36$, so $x=(-2\pm6)/2$; roots are 2 and $-4$. §3.5
3. Both. It is strictly increasing, and for every $y\in\mathbb{R}$, there is a unique real x with $x^3=y$. §3.2
4. $r=\sqrt{(1+3)}=2$. The point is in quadrant III, so $\varphi=4\pi/3$; answer $(2,4\pi/3)$. §3.6
5. Set $y=4x-7$ and solve $x=(y+7)/4$, so $f^{-1}(y)=(y+7)/4$. §3.3
6. $(g\circ f)(x)=3x+2$; $(f\circ g)(x)=3x+10$. The order changes the result. §3.4
7. $f(\{1,3,5\})=\{1,9,25\}$; $\mathrm{Im}(f)$ is a subset of the codomain $\mathbb{N}$, and these are values reached by the selected subset. §3.1
8. $150°=150\pi/180=5\pi/6$. On the unit circle $\cos(5\pi/6)=-\sqrt{3}/2$. §3.5
9. Use domain $[0,\infty)$ and codomain $[0,\infty)$. Then $x^2$ is strictly increasing and every $y\geq0$ has input $\sqrt{y}$. §3.2
10. $2^x=2^4$, hence $x=4$ because $2^x$ is injective for base $2>1$. §3.5
11. $f^{-1}(y)=(y-10)/5$. At $f(2)=20$, $f^{-1}(20)=2$. §3.3
12. $(g\circ f)(x)=\sqrt{(x^2)}=|x|$. Its domain is all $\mathbb{R}$ because f outputs $[0,\infty)$, g's domain. §3.4
13. No. For $x=4$, both $y=2$ and $y=-2$ satisfy $y^2=x$, so one input has two outputs. §3.1
14. $\sin(\pi/3)=\sin(2\cdot\pi/6)=2\sin(\pi/6)\cos(\pi/6)=2\cdot(1/2)\cdot(\sqrt{3}/2)=\sqrt{3}/2$. §3.5
15. $\Delta=2^2-4\cdot3\cdot5=-56<0$, so there are no real roots. §3.5
16. $x=3\cos(\pi/6)=3\sqrt{3}/2$; $y=3\sin(\pi/6)=3/2$. §3.6
17. $1\leq x^2\leq4$, so $x\in[-2,-1]\cup[1,2]$. This is a preimage, not an inverse function. §3.1
18. The degree is 5, so it has at most 5 distinct real roots. §3.5
19. Cosine is restricted to $[0,\pi]$; its range there is $[-1,1]$. §3.5
20. First $(g\circ f)(x)=2(x+1)=2x+2$, then $h(2x+2)=(2x+2)^2$. §3.4
21. When $\cos x=0$, namely $x=\pi/2+k\pi$ for $k\in\mathbb{Z}$. §3.5
22. $\ln(e^3)+\ln(e^2)=3+2=5$. §3.5
23. $r=5$ and the point is on the negative y-axis, so $\varphi=3\pi/2$; answer $(5,3\pi/2)$. §3.6
24. It is both injective and surjective, hence bijective. §3.2

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## Exam-style problems

1. f is strictly increasing on $[0,\infty)$, so it is injective. Every $y\geq 0$ is $f(\sqrt{y})$, so it is onto $[0,\infty)$ and bijective. §3.2
2. Solving $y=x^2$ with $x\geq 0$ gives $f^{-1}(y)=\sqrt{y}$. Then $f^{-1}(f(x))=\sqrt{(x^2)}=x$ for $x\geq 0$. §3.3
3. $(g\circ f)(x)=x^2-1$; its domain is $[0,\infty)$ and its image is $[-1,\infty)$. §3.1 §3.4
4. No. g outputs $[-1,\infty)$, but f accepts only non-negative inputs. For $t\in[0,1)$, $g(t)=t-1$ is negative, so $f(g(t))=f(t-1)$ is undefined. §3.4

## Exam-style problems

1. $\Delta=(-2)^2-4\cdot 1\cdot(-3)=16>0$, so there are two distinct real roots. The formula gives $x=(2\pm 4)/2$, or 3 and $-1$. §3.5
2. $P(x)=(x-3)(x+1)$. Substituting $x=3$ gives $9-6-3=0$. §3.5
3. $x=4\cos(\pi/3)=2$ and $y=4\sin(\pi/3)=2\sqrt{3}$, so $(x,y)=(2,2\sqrt{3})$. §3.6
4. $\sin(\pi/3+\pi/6)=\sin(\pi/3)\cos(\pi/6)+\cos(\pi/3)\sin(\pi/6)=3/4+1/4=1$. The angle is $\pi/2$, where the unit-circle sine is 1. §3.5

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